Open OnDemand for an Established University HPC Community

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UBC-ARC and Sockeye

- UBC ARC supports researchers across all disciplines on campus
- Sockeye is our on-site HPC system managed by ARC and available to campus researchers
  - 15,000+ CPU cores and 100 GPUs
  - Deployed and in operation since fall of 2019
- Supporting over 2,000 researchers across 400 research allocations
- Graphical interfaces for HPC have been a common request across all faculties

Current Architectural Challenges

- Original design and architecture decisions made in 2019 made integrating Open OnDemand more challenging
- User /home directories are read-only on compute nodes
- Compute nodes have very limited outbound network access
- Allocation assignments are separate from central campus LDAP and Shibboleth services

Handling Storage Restrictions

- To limit the impact of heavy job usage on login and file interactions outside of jobs the file system
- Job writing is limited to networked Lustre /scratch and local disks
- OpenOnDemand writes job output and scripts into a user's home directory under an ondemand directory
- Use of the OOD_DATAROOT environment variable allows this to be changed to ensure the writability for jobs

Networking Access Limitations

- Supporting restricted medical data use requirements for our system limited outbound connectivity for compute nodes
- This limits the use of interactive tools to interface with the web for environment setup
- Support team worked to develop a procedure for researchers to install using the OOD shell application
- In the future looking to make interactive applications able to have outbound access to allow seamless environment setup

LDAP User Mapping

- University Enterprise IT provides campus-wide authentication and IAM services
- System tracking of onboarding is done via an ARC managed LDAP endpoint
- Login via Shibboleth was unable to filter users with inactive allocations
- OOD option ‘user_map_cmd’ allowed us to create a local filtering script to verify logins are active users
- Future improvements in our onboarding and allocation process will have Enterprise IT integrations

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